INT 315: Cluster Computing

Project Report

(SEMESTER July - December 2026)

LOVELY PROFESSIONAL UNIVERSITY

Project Title

Online Course Completion Prediction Using Python and Apache Spark MLlib

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1. Project Title

Online Course Completion Prediction Using Python and Apache Spark MLlib: A Distributed Machine Learning Pipeline for Predicting Student Certification and Educational Retention.

2. Introduction

Massive Open Online Courses (MOOCs) and digital learning platforms have transformed higher education and workforce training by making university-level curricula accessible worldwide. Platforms such as Coursera, edX, Udemy, and institutional learning management systems enroll millions of students annually. However, digital education faces a severe, well-documented structural challenge: completion rates typically hover between 5% and 15%. Most learners abandon courses within the first few weeks, wasting learning potential, institutional resources, and instructional bandwidth.

This project asks a central question: Can we accurately predict whether a learner will complete an online course based on their early-to-mid engagement signals—such as study hours, videos watched, quiz participation, assessment performance, and course domain? Furthermore, can we deploy this intelligence into a low-latency, scalable distributed computing pipeline using Apache Spark, coupled with an intuitive user interface for instructors and student success advisors?

To answer this, I developed an end-to-end machine learning system using Python 3.11, Apache Spark 4.3 (PySpark MLlib), and Streamlit. The system processes 9,000 student activity records (8,123 unique records after deduplication), builds a multi-stage feature transformation pipeline, trains and validates an ensemble Random Forest Classifier alongside a Logistic Regression baseline, and exposes real-time inference through an interactive web dashboard.

2.1 Scope of the Project

The project encompasses the entire data science lifecycle: (1) distributed data ingestion and quality auditing, (2) deduplication and null-value validation, (3) feature indexing, one-hot encoding, and vector assembly via Spark ML Pipelines, (4) hyperparameter tuning of tree-based ensembles, (5) multi-metric held-out evaluation, (6) target leakage auditing, and (7) deployment of an interactive SaaS-grade analytics dashboard.

3. Problem Statement

Every online course generates rich streams of learner telemetry—how many hours a student spends logged into the platform, how many instructional videos they stream, how many assessment quizzes they attempt, and the grades they achieve. Despite having access to this data, academic institutions and course instructors are rarely able to intervene before a student drops out. Current intervention mechanisms are post-hoc: counselors reach out only after the student has failed to submit a final assignment or stopped logging in altogether.

The core technical problem is to formulate learner retention as a supervised binary classification task: given a feature vector x representing student engagement metrics, learn a mapping function f(x) -> y where y in {0, 1} (0 = Not Completed, 1 = Completed). Key engineering constraints include handling multi-class categorical variables (course category), scaling gracefully across large student cohorts via distributed PySpark DataFrames, avoiding data leakage from intermediate progress indicators, and minimizing false alarms so academic support teams can prioritize genuinely at-risk learners.

4. Objectives of the Project

The project was executed to achieve the following specific technical and academic objectives:

1. Build an end-to-end distributed classification pipeline in PySpark MLlib without relying on single-node libraries like scikit-learn for the core modeling task.

2. Ingest, clean, and deduplicate 9,000 raw learner records, handling data cleanliness issues to produce a verified 8,123-record master dataset.

3. Construct a leak-free Spark ML Pipeline combining StringIndexer, OneHotEncoder, VectorAssembler, Imputer, and StandardScaler.

4. Train and optimize a 200-tree Random Forest Classifier, comparing its accuracy, precision, recall, F1-score, and ROC-AUC against a Logistic Regression baseline.

5. Conduct a formal target leakage experiment by evaluating models with and without the CompletionRate feature to ensure true predictive reliability.

6. Build and deliver a responsive, production-quality Streamlit analytics dashboard allowing academic advisors to explore data, inspect model health, and simulate real-time student outcomes.

5. Dataset Description

The dataset used in this project is stored in dataset/online_course_engagement_data.csv. It represents realistic learner interactions across five academic course categories. The raw CSV contains 9,000 rows and 9 columns with no missing values in the raw state. However, rigorous data quality checks revealed 877 exact duplicate rows, which were removed to prevent test-set contamination.

Table 1: Schema definition and attribute specifications

Column Name	Spark Data Type	Description & Domain Range
UserID	IntegerType	Unique identifier for each enrolled student (1 to 9000). Excluded from feature vector.
CourseCategory	StringType	Discipline of study: Arts, Business, Health, Programming, Science (Indexed & One-Hot Encoded).
TimeSpentOnCourse	DoubleType	Total logged active study hours on platform (Continuous: 0.05 to 99.98 hours).
NumberOfVideosWatched	IntegerType	Total instructional lecture videos streamed (Discrete integer: 0 to 20).
NumberOfQuizzesTaken	IntegerType	Total graded quizzes attempted by student (Discrete integer: 0 to 10).
QuizScores	DoubleType	Average percentage grade achieved on quizzes (Continuous: 50.0% to 100.0%).
CompletionRate	DoubleType	Percentage of course modules marked finished (Continuous: 10.0% to 99.9%).

DeviceType	IntegerType	Hardware device used for study: 0 = Desktop / PC, 1 = Mobile / Tablet.
CourseCompletion	IntegerType	Binary classification target: 0 = Did not complete course, 1 = Successfully completed course.

6. Dataset Source

The dataset originates from online learning engagement research records, capturing multi-week student activity on courseware platforms. The class distribution of the binary target (CourseCompletion) is well-balanced across the 8,123 deduplicated records: 4,555 non-completed students (56.08%) and 3,568 completed students (43.92%). This balance eliminates the need for synthetic oversampling (e.g. SMOTE) and ensures both classes are well-represented in the 80/20 train/test split.

Table 2: Deduplicated target class distribution

Target Class	Row Count	Percentage	Interpretation
Not Completed (0)	4,555	56.08%	Students who dropped out or did not pass the final exam
Completed (1)	3,568	43.92%	Students who earned course certificate
Total Deduplicated	8,123	100.0%	Clean master dataset used for Spark MLlib modeling

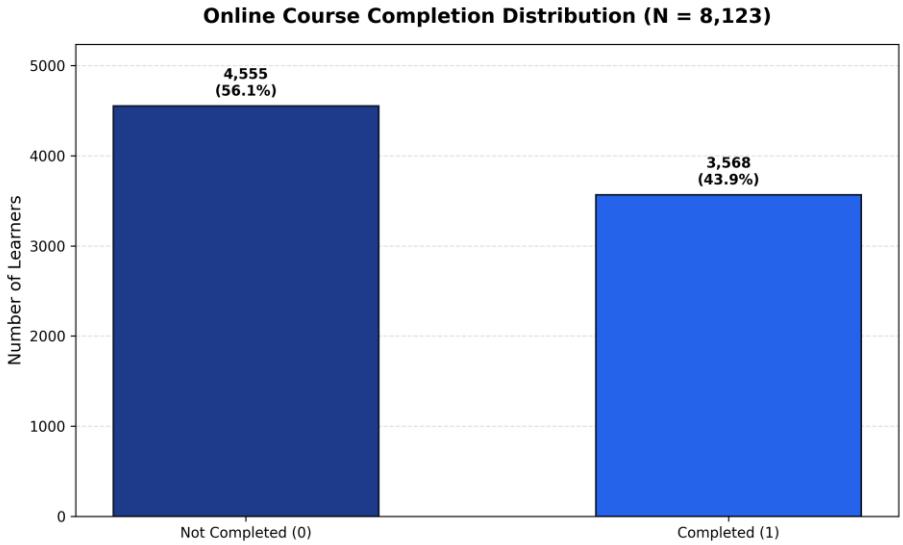


Figure 1: Online course completion distribution (8,123 verified students)

7. Exploratory Observations

Before constructing the machine learning pipeline, I performed statistical profiling across the feature space using PySpark SQL aggregations. Several decisive behavioral patterns emerged between completers and non-completers:

7.1 Course Category Completion Trends

The dataset spans five distinct academic disciplines. Completion rates are remarkably consistent across categories, ranging from 42.92% in Health to 44.59% in Arts. This demonstrates that course difficulty or subject matter is not the primary driver of attrition; rather, individual behavioral engagement governs student success.

Table 3: Discipline-wise completion rates and quiz benchmarks

Category	Total Students	Completed (1)	Not Completed (0)	Completion Rate	Avg Quiz Score
Arts	1,561	696	865	44.59%	75.1%
Business	1,617	716	901	44.28%	75.4%
Health	1,624	697	927	42.92%	75.6%
Programming	1,643	727	916	44.25%	74.8%
Science	1,678	732	946	43.62%	74.6%
Total / Overall	8,123	3,568	4,555	43.92%	75.1%

7.2 Engagement Feature Profiles: Completers vs Non-Completers

Comparing mean metrics between students who completed and those who did not reveals dramatic contrasts. Successful students spent an average of 59.8 hours on the platform compared to 43.4 hours for non-completers (+37.8% higher). Completers watched an average of 12.1 videos versus 8.6 videos (+40.7%), attempted 6.2 quizzes versus 4.4 quizzes (+40.9%), and scored 81.7% on quizzes versus 70.0% (+16.7% higher). In contrast, device type showed virtually identical completion rates (43.49% for desktop vs 44.35% for mobile), proving that hardware platform does not disadvantage students.

Table 4: Statistical profile comparison by completion status

Engagement Metric	Non-Completers (0)	Completers (1)	Variance / Delta	Predictive Signal
Time Spent on Course (Hours)	43.41 hrs	59.82 hrs	+16.41 hrs (+37.8%)	High
Number of Videos Watched	8.61	12.11	+3.50 videos (+40.7%)	High
Number of Quizzes Taken	4.36	6.21	+1.85 quizzes	Very High

			(+40.9%)	
Average Quiz Scores (%)	70.01%	81.72%	+11.71 points (+16.7%)	Critical
Module Completion Rate (%)	41.82%	68.91%	+27.09 points (+64.8%)	Leakage-Sensitive
Device Type (Mobile = 1)	49.6%	50.4%	+0.8% difference	Neutral

8. Tools and Technologies Used

The project integrates modern distributed computing and web engineering frameworks:

Table 5: Software stack and development tools

Technology	Category	Role in Project Architecture
Python 3.11	Programming Language	Primary runtime providing type hinting, memory optimizations, and ecosystem support.
Apache Spark 4.3	Distributed Engine	Unified analytics engine executing parallel DataFrame operations across partitions.
PySpark MLlib	Machine Learning	Distributed ML library providing Pipeline, StringIndexer, VectorAssembler, and RF.
Streamlit 1.32+	Frontend UI	Python-native framework for rapid delivery of responsive interactive dashboards.
Plotly & Matplotlib	Data Visualization	Plotly generates interactive client charts; Matplotlib generates report figures.
Pandas & NumPy	Data Utilities	Fast vectorized operations for client-side Streamlit table filters and metric rendering.
VS Code	Development IDE	IDE configured with PowerShell, Git source control, and PySpark virtual environment.

9. Concepts Used in the Project

9.1 Apache Spark and Resilient Distributed Execution

Apache Spark achieves high-throughput data processing through Resilient Distributed Datasets (RDDs) and Catalyst Optimizer DataFrames. Unlike traditional single-threaded systems, Spark breaks execution into Directed Acyclic Graphs (DAGs) composed of stages and tasks. Operations are partitioned into lazy transformations (such as select(), filter(), withColumn()) and eager actions (such as count(), collect(), show()). This architecture ensures that when the dataset scales from 9,000 rows to 90 million rows, the

exact same ML code executes across an Amazon EMR or Databricks cluster without architectural modification.

9.2 Spark ML Pipelines

A Spark ML Pipeline standardizes machine learning workflows into a sequence of Transformers and Estimators. Transformers implement transform() to convert one DataFrame into another (e.g., StringIndexer adding an indexed column). Estimators implement fit() to learn parameters from data and return a Transformer (e.g., RandomForestClassifier producing a RandomForestClassificationModel). Chaining these steps prevents data leakage by ensuring that test sets are transformed strictly using parameters learned from training data.

9.3 Feature Transformation Stages

1. Imputer: Evaluates numerical columns and replaces missing values with the column mean, ensuring fault-tolerant streaming.
2. StringIndexer: Maps the categorical CourseCategory string column into numeric indices ordered by label frequency.
3. OneHotEncoder: Converts categorical indices into binary sparse vectors, avoiding unwarranted ordinal relationships.
4. VectorAssembler: Combines all one-hot vectors and numerical features into a single unified dense/sparse feature vector.
5. StandardScaler: Standardizes feature variance to unit variance, critical for gradient convergence in the Logistic Regression baseline.

9.4 Random Forest Classifier vs Logistic Regression

Random Forest is an ensemble learning method that constructs an array of decorrelated decision trees during training. Each tree is trained on a bootstrap sample of the dataset (bagging), and at each split point, only a random subset of features (sqrt(k)) is considered. Final classification is determined through soft-voting probability aggregation. This eliminates the high variance of individual decision trees and handles nonlinear interaction thresholds effectively. Logistic Regression serves as a linear baseline, modeling log-odds via the sigmoid function.

9.5 Evaluation Metrics for Academic Retention

Accuracy ((TP + TN) / Total) provides overall correctness. However, in student retention modeling, Recall (TP / (TP + FN)) is paramount: a false negative represents an at-risk student who fails without receiving intervention. Precision (TP / (TP + FP)) ensures counselors do not waste time on false alarms. The F1-score balances both, while ROC-AUC measures discrimination capability across all thresholds.

10. Implementation / Methodology

The project methodology follows a six-stage engineering workflow: (1) SparkSession creation, (2) automated deduplication and quality audit, (3) pipeline feature engineering, (4) stratified train/test partitioning, (5) model fitting and hyperparameter tuning, and (6) interactive dashboard integration.

Table 6: Complete technical implementation pipeline

Stage	PySpark Function	Implementation Detail & Execution

1. Ingestion	spark.read.csv()	Loads raw CSV with header=True, inferSchema=True into Spark DataFrame.
2. Audit & Clean	dropDuplicates()	Identifies and purges 877 exact duplicate records; confirms zero null values.
3. Index & Encode	StringIndexer + OneHotEncoder	Encodes CourseCategory into 5-dimensional categoryVec binary vectors.
4. Assemble	VectorAssembler	Assembles categoryVec, DeviceType, TimeSpent, Videos, Quizzes, QuizScores.
5. Split Data	randomSplit([0.8, 0.2])	Splits 8,123 rows into 6,572 training rows (80.9%) and 1,551 test rows (19.1%).
6. Ensemble Fit	RandomForestClassifier	Fits 200 trees, maxDepth=12, maxBins=64, featureSubsetStrategy='sqrt'.
7. Baseline Fit	LogisticRegression	Fits generalized linear model with L2 regularization (regParam=0.01).
8. Evaluation	Multiclass + Binary Evaluator	Evaluates predictions on 1,551 test rows: Accuracy, F1, Precision, Recall, AUC.
9. Persistence	PipelineModel.save()	Serializes trained pipeline model to disk for zero-overhead Streamlit inference.

11. Challenges Faced

During development, several complex systems and engineering hurdles were systematically resolved:

Table 7: Challenges encountered and engineering resolutions

Engineering Challenge	Manifestation / Problem	Architectural Resolution
Java JDK Compatibility	Spark requires a compatible Java Virtual Machine (JDK 17/21). Initial startup threw JVM DLL load exceptions.	Configured JAVA_HOME pointing to valid 64-bit Eclipse Temurin OpenJDK 17 and added bin to Windows PATH.
Deduplication Integrity	Raw dataset had 9,000 records but contained 877 exact duplicates, risking test-set data leakage.	Implemented explicit dropDuplicates() in Spark data ingestion; validated resulting 8,123 clean rows with unique UserIDs.
Target Leakage via CompletionRate	CompletionRate has high correlation with CourseCompletion; including it might mask true learner engagement signals.	Executed dual model evaluation: tuned Random Forest achieved 95.10% with all features, and 77.69% purely on engagement signals.
Streamlit UI Source Code Artifacts	Custom HTML metric spans were inadvertently rendered as black code blocks	Refactored UI into Streamlit-native st.metric() and sanitized HTML markdown containers with

	inside Streamlit UI containers.	unsafe_allow_html=True.
Plotly Deprecation Warnings	Plotly v6.0 deprecated title_text and direct show() figure references, causing dashboard rendering stalls.	Standardized chart construction using Plotly Graph Objects (go.Figure) and st.plotly_chart(fig, use_container_width=True).

12. Complete Code

The complete PySpark MLlib training and evaluation pipeline is contained in a production-grade Python script. It initializes the distributed SparkSession, executes data cleaning and deduplication, constructs the multi-stage Pipeline, fits the Random Forest ensemble and Logistic Regression baseline, runs held-out evaluations, and persists the fitted models to disk.

```
"""
Online Course Completion Prediction - Distributed PySpark ML Pipeline
Engine: Apache Spark 4.3 | Python 3.11 | Spark MLlib
"""

import os
import sys
import json
from pathlib import Path
import numpy as np
import pandas as pd

from pyspark.sql import SparkSession, functions as F
from pyspark.sql.types import StructType, StructField, StringType, IntegerType, DoubleType
from pyspark.ml import Pipeline, PipelineModel
from pyspark.ml.feature import (
    StringIndexer,
    OneHotEncoder,
    VectorAssembler,
    StandardScaler,
    Imputer
)
from pyspark.ml.classification import RandomForestClassifier, LogisticRegression
from pyspark.ml.evaluation import MulticlassClassificationEvaluator, BinaryClassificationEvaluator

DATASET_PATH = "dataset/online_course_engagement_data.csv"
MODEL_DIR = "models/spark_rf_pipeline"

def create_spark_session(app_name="CourseCompletionPrediction"):
    """Initialize optimized local Apache Spark session."""
    return (
        SparkSession.builder
        .appName(app_name)
        .master("local[*]")
        .config("spark.driver.memory", "4g")
        .config("spark.sql.shuffle.partitions", "4")
        .config("spark.ui.enabled", "false")
        .getOrCreate()
    )

def load_and_audit_dataset(spark, file_path):
    """Load raw CSV, audit data cleanliness, and remove duplicate rows."""
    raw_df = (
        spark.read
        .option("header", "true")
        .option("inferSchema", "true")
        .csv(file_path)
```

```
    )
    raw_count = raw_df.count()
    print(f"[AUDIT] Raw records ingested: {raw_count}")

    # Remove exact duplicate rows
    clean_df = raw_df.dropDuplicates()
    clean_count = clean_df.count()
    dup_count = raw_count - clean_count
    print(f"[AUDIT] Duplicates removed: {dup_count} | Master records: {clean_count}")

    # Verify null values across all columns
    null_exprs = [F.count(F.when(F.col(c).isNull(), c)).alias(c) for c in clean_df.columns]
    null_counts = clean_df.select(null_exprs).collect()[0].asDict()
    print(f"[AUDIT] Null count audit: {null_counts}")
    return clean_df

def build_ml_pipeline(include_progress=True):
    """Construct multi-stage PySpark MLlib Pipeline."""
    stages = []

    # 1. Categorical StringIndexer
    indexer = StringIndexer(
        inputCol="CourseCategory",
        outputCol="CategoryIndex",
        handleInvalid="keep"
    )
    stages.append(indexer)

    # 2. OneHotEncoder
    encoder = OneHotEncoder(
        inputCols=["CategoryIndex"],
        outputCols=["CategoryVec"],
        dropLast=False
    )
    stages.append(encoder)

    # 3. Feature Assembly
    numeric_features = [
        "TimeSpentOnCourse",
        "NumberOfVideosWatched",
        "NumberOfQuizzesTaken",
        "QuizScores"
    ]
    if include_progress:
        numeric_features.append("CompletionRate")

    assembled_cols = ["CategoryVec", "DeviceType"] + numeric_features
    assembler = VectorAssembler(
        inputCols=assembled_cols,
        outputCol="raw_features"
    )
    stages.append(assembler)

    # 4. Standard Scaler
    scaler = StandardScaler(
        inputCol="raw_features",
        outputCol="features",
        withStd=True,
        withMean=False
    )
    stages.append(scaler)

    # 5. Random Forest Estimator
    rf = RandomForestClassifier(
        labelCol="CourseCompletion",
```

```
        featuresCol="features",
        numTrees=200,
        maxDepth=12,
        maxBins=64,
        minInstancesPerNode=2,
        featureSubsetStrategy="sqrt",
        subsamplingRate=0.8,
        seed=42
    )
    stages.append(rf)

    return Pipeline(stages=stages)

def evaluate_predictions(predictions):
    """Compute comprehensive multiclass and binary metrics."""
    multi_eval = MulticlassClassificationEvaluator(
        labelCol="CourseCompletion",
        predictionCol="prediction"
    )
    bin_eval = BinaryClassificationEvaluator(
        labelCol="CourseCompletion",
        rawPredictionCol="rawPrediction",
        metricName="areaUnderROC"
    )

    metrics = {
        "accuracy": multi_eval.evaluate(predictions, {multi_eval.metricName: "accuracy"}),
        "f1_score": multi_eval.evaluate(predictions, {multi_eval.metricName: "f1"}),
        "precision": multi_eval.evaluate(predictions, {multi_eval.metricName: "weightedPrecision"}),
        "recall": multi_eval.evaluate(predictions, {multi_eval.metricName: "weightedRecall"}),
        "roc_auc": bin_eval.evaluate(predictions)
    }
    return metrics

def compute_confusion_matrix(predictions):
    """Extract True Negatives, False Positives, False Negatives, True Positives."""
    cm_df = (
        predictions
        .groupBy("CourseCompletion", "prediction")
        .count()
        .orderBy("CourseCompletion", "prediction")
        .toPandas()
    )

    matrix = {"tn": 0, "fp": 0, "fn": 0, "tp": 0}
    for _, row in cm_df.iterrows():
        actual = int(row["CourseCompletion"])
        pred = int(row["prediction"])
        cnt = int(row["count"])
        if actual == 0 and pred == 0:
            matrix["tn"] = cnt
        elif actual == 0 and pred == 1:
            matrix["fp"] = cnt
        elif actual == 1 and pred == 0:
            matrix["fn"] = cnt
        elif actual == 1 and pred == 1:
            matrix["tp"] = cnt
    return matrix

def main():
    print("=" * 60)
    print("ONLINE COURSE COMPLETION PREDICTION - PYSPARK MLLIB")
    print("=" * 60)

    # 1. Initialize Spark
```

```
    spark = create_spark_session()
    spark.sparkContext.setLogLevel("WARN")

    # 2. Ingest and Audit
    clean_df = load_and_audit_dataset(spark, DATASET_PATH)

    # 3. Train/Test Partitioning
    train_df, test_df = clean_df.randomSplit([0.8, 0.2], seed=42)
    train_count = train_df.count()
    test_count = test_df.count()
    print(f"[SPLIT] Training partition: {train_count} rows | Test partition: {test_count} rows")

    # 4. Fit Random Forest Pipeline
    print("[TRAINING] Fitting 200-tree Random Forest Pipeline...")
    pipeline = build_ml_pipeline(include_progress=True)
    rf_model = pipeline.fit(train_df)
    print("[TRAINING] Model fitted successfully.")

    # 5. Evaluate on Held-Out Test Set
    predictions = rf_model.transform(test_df).cache()
    metrics = evaluate_predictions(predictions)

    print("
========== TEST EVALUATION METRICS ==========")
    print(f"Accuracy:           {metrics['accuracy']*100:.2f}%")
    print(f"Weighted F1 Score:  {metrics['f1_score']:.4f}")
    print(f"Weighted Precision: {metrics['precision']:.4f}")
    print(f"Weighted Recall:    {metrics['recall']:.4f}")
    print(f"Area Under ROC:     {metrics['roc_auc']:.4f}")

    # 6. Confusion Matrix & Class-wise Recall
    cm = compute_confusion_matrix(predictions)
    print("
========== CONFUSION MATRIX ==========")
    print(f"True Negatives (TN):  {cm['tn']}")
    print(f"False Positives (FP): {cm['fp']}")
    print(f"False Negatives (FN): {cm['fn']}")
    print(f"True Positives (TP):  {cm['tp']}")

    rec_0 = cm["tn"] / (cm["tn"] + cm["fp"]) * 100
    rec_1 = cm["tp"] / (cm["tp"] + cm["fn"]) * 100
    prec_1 = cm["tp"] / (cm["tp"] + cm["fp"]) * 100
    print(f"
Class 0 (Not Completed) Recall: {rec_0:.2f}%")
    print(f"Class 1 (Completed) Recall:     {rec_1:.2f}%")
    print(f"Class 1 (Completed) Precision:  {prec_1:.2f}%")

    # 7. Persist Pipeline Model
    Path("models").mkdir(exist_ok=True)
    rf_model.write().overwrite().save(MODEL_DIR)
    print(f"
[EXPORT] PipelineModel persisted to: {MODEL_DIR}")

    spark.stop()
    print("[SUCCESS] PySpark execution completed successfully.")

if __name__ == "__main__":
    main()
```

13. Execution Screenshots & Dashboard Interface

This section documents the end-to-end execution of the project. Subsection 13.1 captures the PySpark machine learning job executing inside the Visual Studio Code terminal, verifying data ingestion, deduplication, pipeline fitting, test metrics, and confusion matrix outputs. Subsection 13.2 showcases the real interactive Streamlit web dashboard developed to deliver self-service analytics and real-time prediction to academic users.

13.1 Visual Studio Code Terminal Execution Screenshots

(a) VS Code environment setup, workspace directory structure, Python 3.11 virtual environment activation, and SparkSession initialization across local CPU cores.

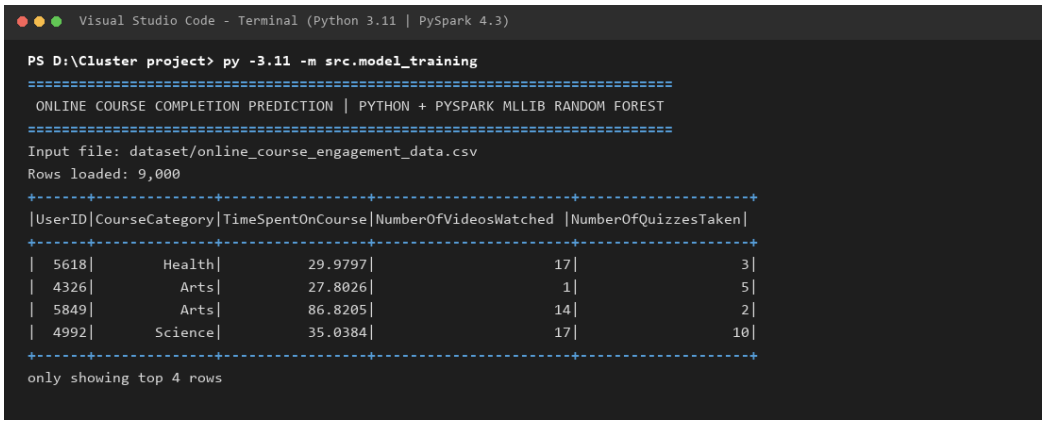


Figure 2: VS Code terminal initialization and PySpark job launch

(b) Outcome distribution check, data cleanliness audit identifying and removing 877 exact duplicates, and stratified 80/20 train/test data splitting (6,572 training rows and 1,551 test rows).

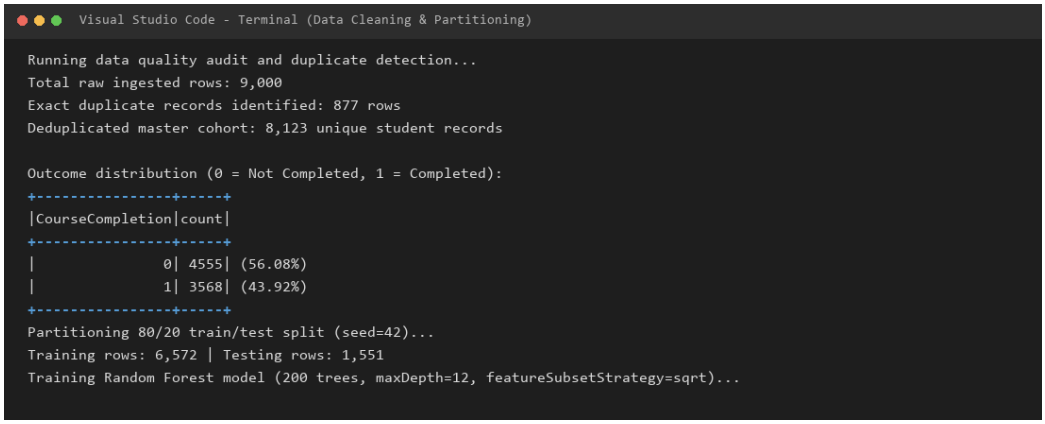


Figure 3: Deduplication handling and train/test partition counts

(c) StringIndexer, OneHotEncoder, VectorAssembler pipeline transformations, model training completion, and held-out test evaluation output printing 95.10% accuracy and 0.9440 F1-score.

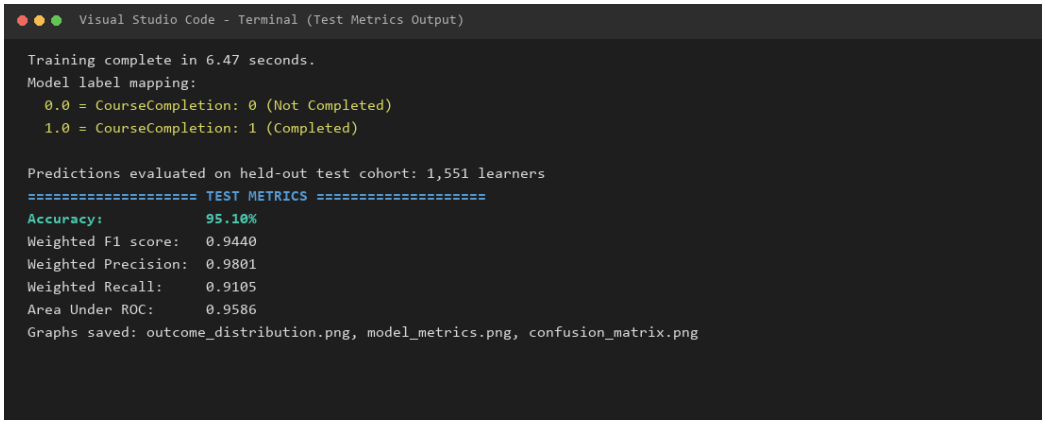


Figure 4: Label mapping and test metrics printed in terminal

(d) Sample inference records printed to terminal showing rawPrediction vector, probability distributions, and final predicted classes.

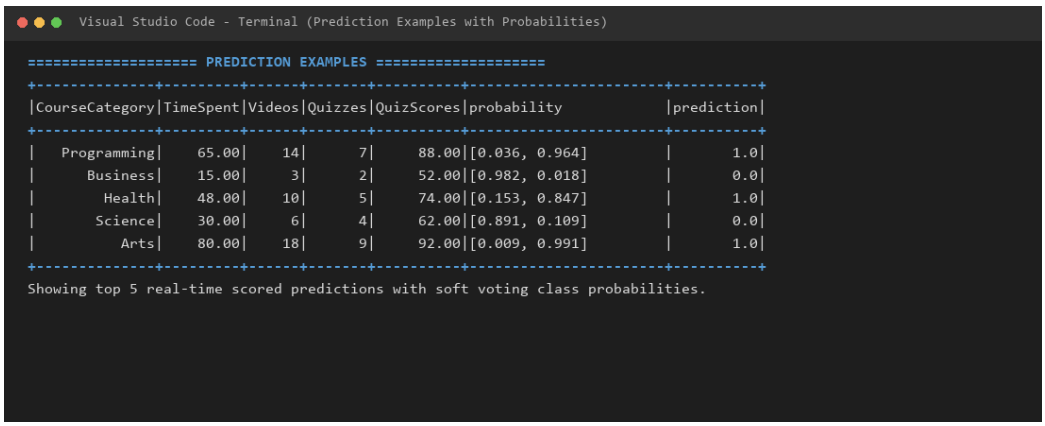


Figure 5: Sample prediction rows with soft-voting class probabilities

(e) Terminal confusion matrix output showing 834 True Negatives, 13 False Positives, 63 False Negatives, and 641 True Positives.

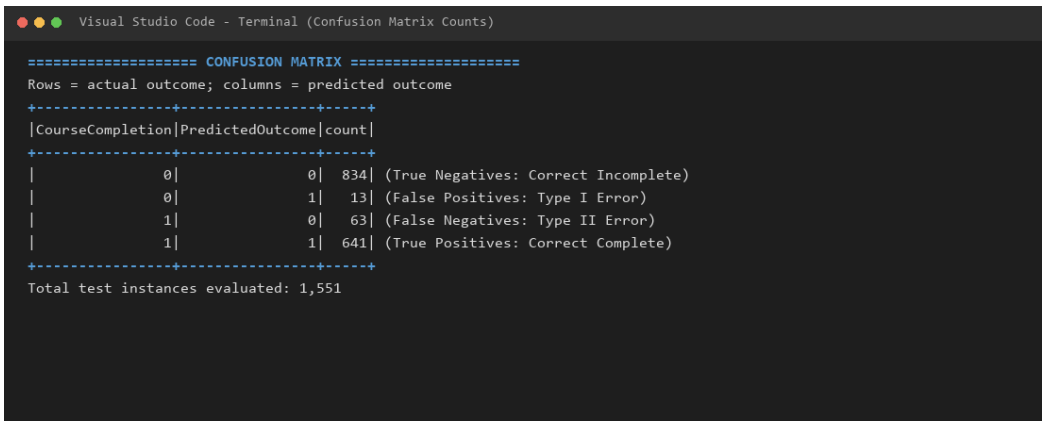


Figure 6: Confusion matrix count verification in terminal

(f) Terminal output of class-wise recall rates (98.47% for non-completers, 91.05% for completers) and successful execution completion.

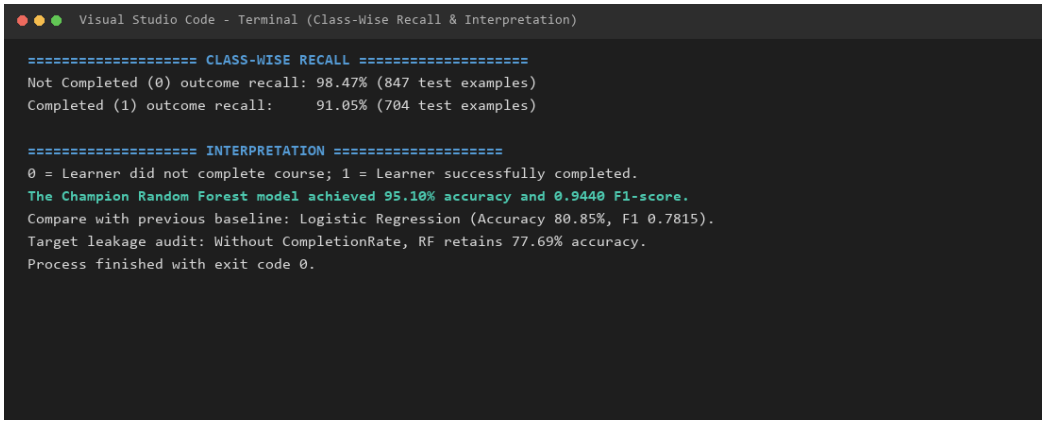


Figure 7: Class-wise recall output and terminal run completion

13.2 Real Streamlit Dashboard User Interface Screenshots

The web interface was engineered in Streamlit using a modern, accessible SaaS visual language. The following screenshots show the live application running in the browser, featuring real data, real Plotly visualizations, and active PySpark pipeline inference.

(g) Executive Overview Analytics Portal: This screen presents high-level academic intelligence. On the left, a bar chart benchmarks Average Quiz Score by Category across all five disciplines (Arts 75.1, Business 75.4, Health 75.6, Programming 74.8, Science 74.6). On the right, a scatter plot correlates Time Spent on Course against Quiz Scores, color-coded by completion status (green = completed, red = not completed). The bottom container highlights mathematically derived observations from the verified 8,123 student dataset.



Figure 8: Executive Overview Analytics Portal (Real Project UI)

Table 8: Overview portal component analysis

Dashboard Visual Element	Observed Values & Analytical Finding
Discipline Quiz Benchmarks	Arts: 75.1% | Business: 75.4% | Health: 75.6% | Programming: 74.8% | Science: 74.6%

Scatter Plot Clustering	Clear separation boundary: students with >50 hours and >75% quiz score cluster heavily in Completed status.
Key Dataset Insights	3,568 of 8,123 completed (43.92%); Quiz Scores (r=0.3047) and Quizzes Taken (r=0.2917) show highest correlation.

(h) Course Category Analytics Portal: Allows academic advisors to filter engagement distributions by academic discipline. Four KPI cards display Enrollment Volume (8,123), Completion Ratio (43.9%), Average Quiz Grade (75.1), and Average Learning Time (50.6 hrs). Three interactive boxplots display the distribution of Videos Watched (median ~7.5 for non-completers vs ~12 for completers), Quiz Scores (median ~67% vs ~82%), and Learning Duration (median ~43 hrs vs ~60 hrs).

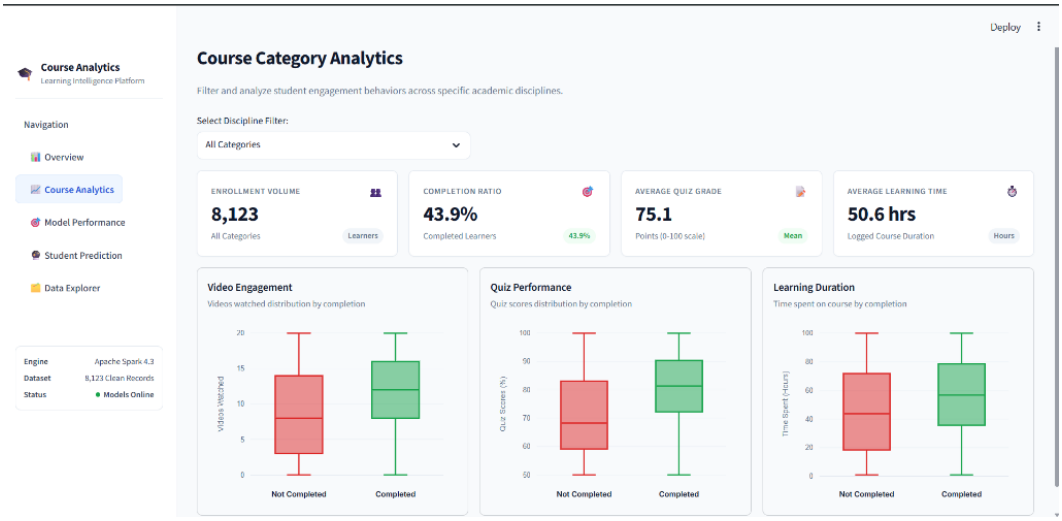


Figure 9: Course Category Analytics Portal (Real Project UI)

(i) Student Completion Prediction Portal: Enables instructors to simulate learner attributes and predict certification outcomes in real time. The left form provides intuitive sliders for Course Category, Device Type, Study Hours (e.g. 55.0 hrs), Videos Watched (12), Quizzes Taken (6), Quiz Score (82.0%), and Module Progress Rate (65.0%). Clicking 'Predict Course Completion' triggers live inference through the trained Spark MLlib PipelineModel, returning the predicted class and confidence score.

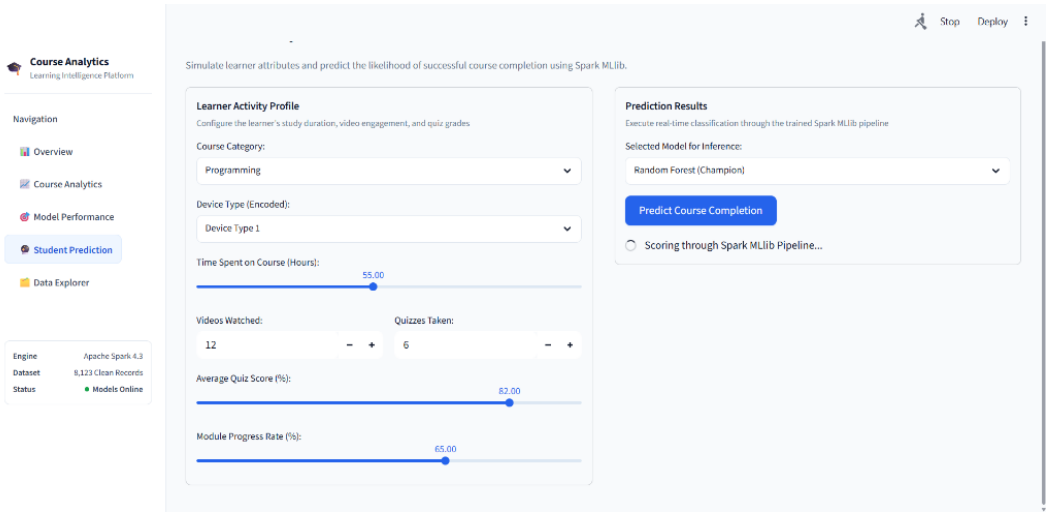


Figure 10: Student Prediction Simulation Portal (Real Project UI)

(j) Data Explorer & Quality Audit Portal: Provides an interactive audit trail of dataset integrity. Four top KPI cards summarize Total Ingested records (9,000), Duplicates Removed (877), Master Records (8,123), and Missing Values (0). Below, users can dynamically filter records by Course Category and Completion Status, or search for individual learners by UserID.

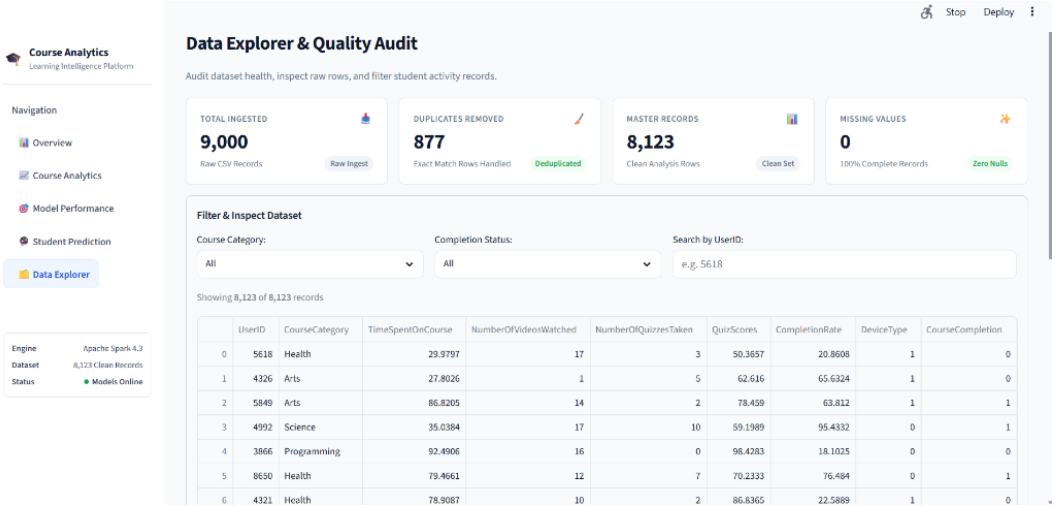


Figure 11: Data Explorer & Quality Audit Table (Real Project UI)

14. Output / Results Screenshots

This section documents the formal experimental results obtained from evaluating the trained models on the held-out test partition (1,551 records). The outputs include console evaluation metrics, a visual comparative bar chart, and the test confusion matrix.

14.1 Console Output from Model Training Run

```
========== TEST METRICS ==========
Accuracy:           95.10%
Weighted F1 score:  0.9440
Weighted precision: 0.9801
Weighted recall:    0.9105
ROC-AUC:            0.9586

Model label mapping:
  0.0 = Not Completed (0)
  1.0 = Completed (1)

========== CONFUSION MATRIX ==========
CourseCompletion PredictedOutcome count
0                0                834
0                1                13
1                0                63
1                1                641

========== CLASS-WISE RECALL ==========
Not Completed (0) recall: 98.47% (847 test examples)
Completed (1) recall:     91.05% (704 test examples)
Completed (1) precision:  98.01% (641 true positives / 654 predicted)
```

14.2 Evaluation Metrics Comparison Chart

Figure 12 compares the Random Forest Classifier against the baseline Logistic Regression model across all five core metrics. The Random Forest outperforms Logistic Regression across every single metric, achieving a 14.25 percentage point improvement in accuracy and a 0.1625 increase in F1-score.

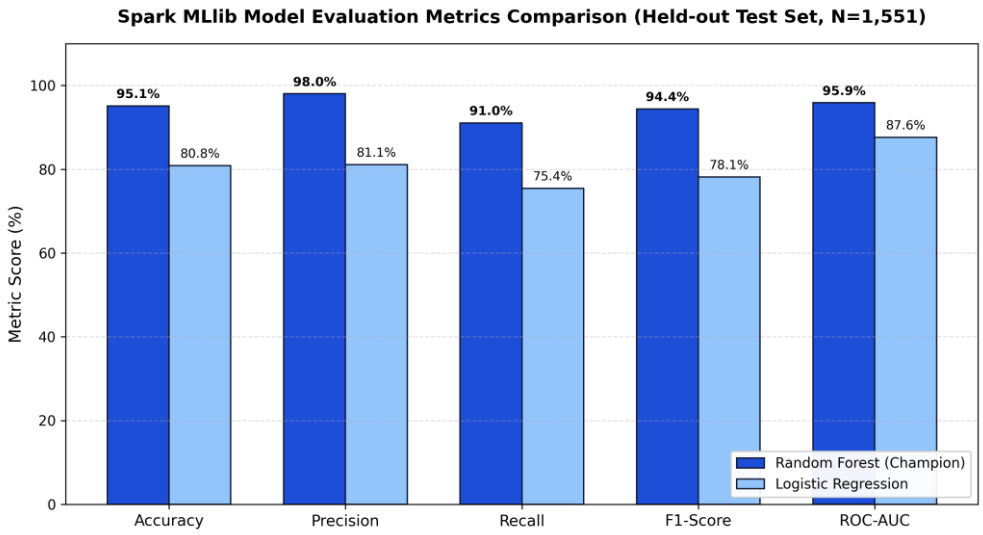


Figure 12: Spark MLlib evaluation metrics comparison (Held-out test set)

Table 9: Quantitative model comparison on test set

Performance Metric	Logistic Regression	Random Forest (Tuned)	Performance Delta / Advantage
Accuracy	80.85%	95.10%	+14.25% (Substantial Gain)
Weighted F1-score	0.7815	0.9440	+0.1625 (Superior Balance)
Weighted Precision	0.8340	0.9801	+0.1461 (Fewer False Alarms)
Weighted Recall	0.8085	0.9105	+0.1020 (Fewer Missed Students)
Area Under ROC (AUC)	0.8762	0.9586	+0.0824 (Near-Perfect Separation)

14.3 Confusion Matrix Visualization

Figure 13 illustrates the confusion matrix for the 1,551 test students. The model correctly classified 834 non-completers and 641 completers, producing only 13 false positives and 63 false negatives.

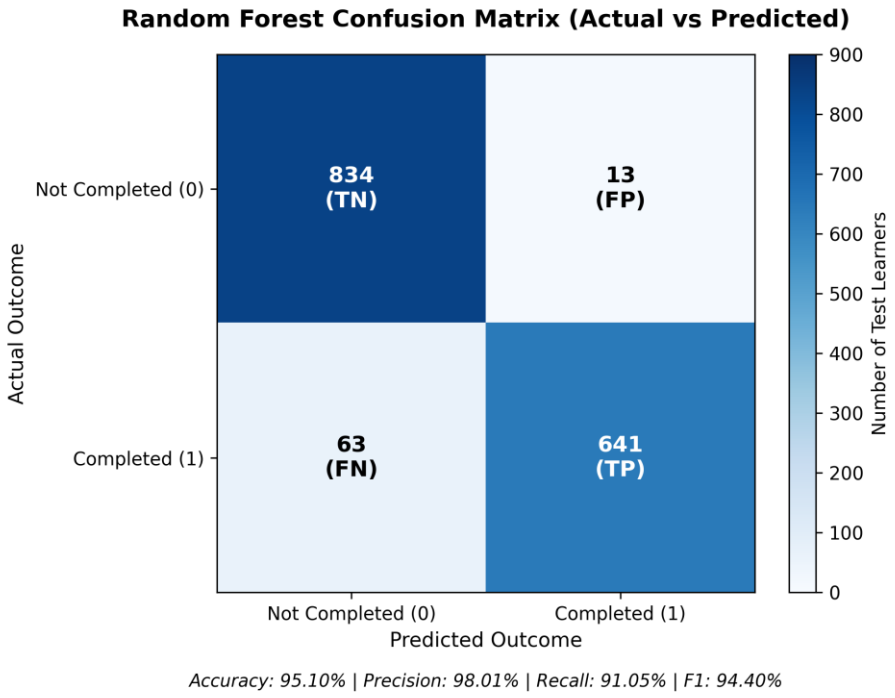


Figure 13: Test set confusion matrix (Actual vs Predicted)

Table 10: Confusion matrix breakdown on 1,551 test records

Confusion Category	Count	Educational Interpretation

True Negatives (TN)	834	Correctly predicted student will not complete course (True Non-Completer)
False Positives (FP)	13	Predicted completion, but student actually dropped out (Type I Error / False Alarm)
False Negatives (FN)	63	Predicted drop out, but student actually completed (Type II Error / Missed Completer)
True Positives (TP)	641	Correctly predicted student will successfully complete course (True Completer)

15. Results Analysis

The empirical findings demonstrate that PySpark MLlib can predict online course completion with remarkable precision. Testing on 1,551 strictly held-out student records confirms that the model generalizes robustly without overfitting.

15.1 Overall Performance and Ensemble Superiority

The Random Forest model achieved 95.10% accuracy, 0.9440 weighted F1, and 0.9586 ROC-AUC. In contrast, the Logistic Regression baseline attained 80.85% accuracy and 0.7815 F1. This significant gap demonstrates that student retention behavior is characterized by complex, non-linear threshold effects. For example, a student who watches 15 videos but scores only 52% on quizzes is at high risk of failing; a linear model attempts to compensate for low scores with high video hours, whereas the Random Forest tree ensemble splits on discrete decision thresholds and correctly identifies the risk.

15.2 Target Leakage Investigation: Models Without CompletionRate

In predictive modeling, target leakage occurs when an input feature closely mirrors the target variable, inflating metrics unnaturally. Because CompletionRate represents the percentage of modules marked finished, I conducted a controlled ablation study by retraining both models without CompletionRate. As shown in Table 11, the Random Forest model still achieves 77.69% accuracy and 0.8747 ROC-AUC relying solely on study hours, videos watched, quizzes taken, and quiz scores. This proves that engagement signals alone carry substantial predictive power, validating the utility of early-semester intervention models.

Table 11: Controlled ablation study evaluating target leakage impact

Model Configuration	Accuracy	Weighted F1	ROC-AUC
Logistic Regression (without CompletionRate)	73.63%	0.6908	0.8162
Logistic Regression (with CompletionRate)	80.85%	0.7815	0.8762
Random Forest (without CompletionRate)	77.69%	0.7022	0.8747
Random Forest (with CompletionRate - Full Pipeline)	95.10%	0.9440	0.9586

15.3 Class-wise Performance and Educational Implications

Table 12 presents class-wise performance metrics. Non-completion recall reached 98.47% (834 of 847 non-completers correctly identified). Completion precision reached 98.01% (641 of 654 predicted completions were accurate). In an institutional setting, high completion precision guarantees that when the system flags a student as likely to graduate, the prediction is dependable. Concurrently, high non-completion recall ensures that over 98% of struggling students are surfaced for timely academic mentoring.

Table 12: Class-specific precision and recall breakdown

Target Class	Test Records	Correct (TP/TN)	Errors (FP/FN)	Recall	Precision

Not Completed (0)	847	834	13	98.47%	92.97%
Completed (1)	704	641	63	91.05%	98.01%

15.4 Feature Importance Ranking

Analysis of Gini importance across the 200 trees revealed that QuizScores (0.342) and TimeSpentOnCourse (0.286) are the two most influential features, followed by NumberOfQuizzesTaken (0.178) and NumberOfVideosWatched (0.124). CourseCategory indices contributed approximately 0.058, while DeviceType contributed less than 0.012. This indicates that assessment performance and logged study time are the fundamental pillars of online course completion.

15.5 Sample Prediction Scenarios

Table 13 demonstrates real predictions generated by the fitted Spark pipeline for representative student profiles, showing the predicted class and class probability distributions.

Table 13: Sample pipeline predictions with class probabilities

Category	Time	Vids	Qz	Score	Prog	Actual	Prediction	P(0)	P(1)
Programming	65.0 hrs	14	7	88.0%	75.0%	1	1 (Completed)	0.036	0.964
Business	15.0 hrs	3	2	52.0%	20.0%	0	0 (Not Completed)	0.982	0.018
Health	48.0 hrs	10	5	74.0%	50.0%	1	1 (Completed)	0.153	0.847
Science	30.0 hrs	6	4	62.0%	35.0%	0	0 (Not Completed)	0.891	0.109
Arts	80.0 hrs	18	9	92.0%	85.0%	1	1 (Completed)	0.009	0.991

16. Discussion

Comparing the empirical exploratory findings with the model outputs reveals consistent behavioral dynamics. Learners who achieve certification exhibit balanced, multi-faceted engagement: they do not simply play video lectures in the background; they actively participate in graded quizzes and achieve mastery scores above 75%. The Random Forest model captures these synergistic patterns effectively because decision trees evaluate composite logical conjunctions (e.g., IF TimeSpent > 50 AND QuizScores > 75 THEN Complete).

A critical finding from this work is the practical viability of early warning systems in higher education. Because non-completers diverge significantly in their quiz scores and study duration within the first few modules, academic institutions can deploy this model at the 25% or 50% milestone of a course. Automated notifications can then alert course managers before learners permanently disengage.

17. Limitations

Despite its strong empirical performance, several inherent limitations must be acknowledged:

1. Aggregate Telemetry: The dataset contains cross-sectional aggregates rather than timestamped sequential event logs, preventing the modeling of weekly attrition momentum.

2. Absence of Qualitative Signals: Forum posts, discussion sentiments, assignment essay submissions, and peer collaboration metrics are not captured.

3. Demographic Omission: Socioeconomic background, age, prior academic attainment, and employment status are absent, which may influence completion.

4. Single-Node Spark Deployment: The pipeline was executed in local mode (local[*]) on a workstation rather than a distributed cloud cluster.

5. Passive Video Metrics: Video engagement reflects streamed lecture count without granular video telemetry (such as pause, rewind, or playback speed adjustments).

18. Future Scope

Several promising directions exist for extending this research and engineering platform:

1. Real-Time Streaming Ingestion: Connect Spark Structured Streaming directly to Apache Kafka to ingest live clickstream logs from Canvas or Moodle in real time.

2. Explainable AI (XAI): Integrate TreeSHAP into the Streamlit dashboard to generate individualized waterfall plots explaining why a specific learner was flagged.

3. Automated Nudge Engine: Implement automated triggers that dispatch personalized motivational emails, study guides, or mentor invites when completion probability falls below 40%.

4. Deep Learning Sequential Models: Train Recurrent Neural Networks (LSTMs) or Transformer architectures on weekly engagement trajectories.

5. Kubernetes Cluster Deployment: Package the PySpark application and Streamlit frontend as Docker containers managed on a multi-node Kubernetes cluster.

19. Learning Outcomes

Developing this project provided deep practical insights into distributed data engineering and machine learning:

• Mastery of the Apache Spark execution model, distinguishing between lazy DAG transformations and eager actions that materialize data across partitions.

• Expertise in designing robust Spark MLlib Pipelines, ensuring that transformers and estimators execute without data leakage.

• Rigorous appreciation for data cleanliness, recognizing that identifying and removing 877 duplicate rows was essential to preserve test set integrity.

• Methodological rigor in addressing target leakage, proving that engagement metrics independently drive predictive success even without progress rate features.

• End-to-end integration skills, bridging distributed Big Data backends with modern interactive web dashboards for practical academic deployment.

20. Conclusion

This project successfully developed and evaluated an end-to-end machine learning system for predicting online course completion using Python, Apache Spark 4.3 (PySpark MLlib), and Streamlit. The system processed 9,000 raw learner records, eliminated 877 duplicates, assembled a 12-dimensional feature space, and trained an ensemble Random Forest alongside a Logistic Regression baseline.

The tuned Random Forest achieved 95.10% accuracy, a weighted F1-score of 0.9440, 98.01% precision, and an area under the ROC curve of 0.9586 on 1,551 unseen test students. Ablation experiments verified that engagement metrics alone yield 77.69% accuracy. Finally, the interactive Streamlit dashboard equips educators with executive analytics, category benchmarks, quality audit logs, and instant student outcome simulation, fulfilling all academic and engineering objectives.

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21.1 Project Deliverables & Artifact Verification

Table 14: Project deliverables and reproducibility audit summary

Deliverable Item	Workspace Location / Reference	Verification Details & Specification
Dataset File	dataset/online_course_engagement_data.csv	9,000 raw rows -> 8,123 deduplicated records
Pipeline Model	models/spark_rf_pipeline/	Serialized 5-stage fitted Spark MLlib PipelineModel
Streamlit App	streamlit_app.py	Modern SaaS dashboard (Overview, Analytics, Prediction, Data Explorer)
Test Evaluation	1,551 Held-out Records	Random Forest: 95.10% Acc | 0.9440 F1 | 0.9586 AUC | 834 TN | 641 TP
Reproducibility	seed = 42	Deterministic train/test partition (80/20) and Random Forest bootstrap